Hierarchical-Instance Contrastive Learning for Minority Detection on Imbalanced Medical Datasets

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***Abstract*— Deep learning methods are often hampered by issues such as data imbalance and data-hungry. In medical imaging, malignant or rare diseases are frequently of minority classes in the dataset, featured by diversified distribution. Besides that, insufficient labels and unseen cases also present conundrums for training on the minority classes. To confront the stated problems, we propose a novel Hierarchical-instance Contrastive Learning (HCLe) method for minority detection by only involving data from the majority class in the training stage. To tackle inconsistent intra-class distribution in majority classes, our method introduces two branches, where the first branch employs an auto-encoder network augmented with three constraint functions to effectively extract image-level features, and the second branch designs a novel contrastive learning network by taking into account the consistency of features among hierarchical samples from majority classes. The proposed method is further refined with a diverse mini-batch strategy, enabling the identification of minority classes under multiple conditions. Extensive experiments have been conducted to evaluate the proposed method on three datasets of different diseases and modalities. The experimental results show that the proposed method outperforms the state-of-the-art methods.**

***Index Terms*— Data imbalance, minority detection, hierarchical instance, medical image analysis.**

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I. INTRODUCTION

TRAINING a deep neural network (DNN) to perform well requires a relatively balanced dataset, and imbalanced ones may lead to a biased model towards the majority class while ignoring the minority class [1]. In medical data, samples in some specific or severe diseases are often insufficient, even though these diseases are usually seen as of scientific interest and essence [2].

Conventional methods for dealing with data imbalance are re-sampling [3], [4], [5], [6], re-weighting [7], [8], [9], [10], and synthesizing data [11] or labels [12]. Re-sampling is to increase the number of minority samples. Re-weighting adjusts the weights between the minority and majority classes, which aims to re-balance these imbalanced data. However, most of these conventional methods focus on re-balancing samples among classes but still lack samples of minority classes, which is inappropriate for clinical practice due to 1) Limited contribution from rare cases of clinical significance in minority classes [13]; 2) Poor performance on learning diverse cases. Even after rebalancing, it is still unlikely to increase the underlying information of minority classes. Certain minority classes (e.g., rare or malignant diseases) have a limited number of samples and are mostly the class of interest and essence [14], some of which are even unseen in some medical institutions. Thus it is difficult to train a network based on such limited data. Apart from the rare case issue, the diversity of cases also poses difficulties for learning. In terms of minority classes, besides the sample size issues, these cases in a minority class (e.g., malignant) are accompanied by distribution diversity. For instance, certain cancers vary considerably in terms of shape, color, and location distributions [15]. Also, the consistency of the majority class fails to be guaranteed in numerous imbalanced medical scenarios. The majority class mainly contains multiple sub-classes in clinical practice (e.g., several sub-classes in a benign class). In this way, conventional methods distinguish sub-classes of the majority class and minority class by learning their diverse features implicitly. Due to the high diversity and lack of numbers in the imbalanced data, the decision boundaries could be biased toward the majority class, and thus it is challenging to distinguish minority classes.

In this paper, we propose a novel method ***H**ierarchical-instance **C**ontrastive **Le**arning* (**HCLe**) for minority detection. The HCLe requires only the majority class in the training process to avoid training on the rare and diverse minority. Unlike conventional methods of learning biased features, it learns consistent features of the majority class. Due to the influence of diverse cases, our proposed method has two branches. Firstly, we introduce an image reconstruction branch augmented with multi-constraint functions (i.e., SSIM, GMS, and adversarial losses) to enhance the feature extraction ability. This image reconstruction branch is derived from conventional encoder-decoder networks. Secondly, to address the inconsistency issue of the majority class (e.g., high intra-class variation [16]), our method uses a novel hierarchical-instance contrastive learning branch to improve the consistency of feature representation among majority samples. This branch learns intra-class consistent features by sampling hierarchical instances from sub-classes of the majority class as contrastive pairs. Our hierarchical-instance contrastive learning network shares some similarities with multi-instance contrastive learning (MICLe) [17] which also uses multiple instances for contrastive learning. However, we emphasize to a greater extent on the hierarchical relationship among the instances, which can cope with the imbalanced data in a better way. These two branches share the same feature embedding, thus considering both feature representations and the similarity within the majority class. In the inference stage, the minority class (e.g., certain malignant diseases) is detected because its distribution differs from the majority class. For instance, melanoma (a skin cancer) has asymmetry, blurred irregularity, and uneven color distribution, different from benign moles [18]. Although our method is inspired by anomaly detection methods [19], [20], [21], [22], they are not designed for imbalanced data and can only address issue #1 (i.e., limited contribution from rare cases of clinical significance in minority classes).

We evaluate our method with conventional and anomaly detection methods on three imbalanced medical datasets with different modalities. In brief, our main contributions are summarised as follows:

- We propose a new method HCLe to address data imbalance, which only requires the majority class in the training process and learns the majority class' consistent features.
- The proposed HCLe has two branches, image reconstruction and hierarchical-instance contrastive learning. These two branches share the same encoder for mining the complementary information in both good image-level features and intra-class similarity features.
- For the hierarchical-instance contrastive learning branch, a novel strategy is introduced by minimizing the distance among hierarchical samples of the majority class to keep the intra-class feature similar. By designing a mini-batch of contrast inputs, this method identifies minority classes (e.g., certain malignant diseases) underlying the multiple cases condition. Unlike many methods that are only available for a single case to identify the malignant disease, our approach can be applied for more complex cases.
- Our method is compared with conventional imbalanced methods and anomaly detection methods, is validated on three imbalanced medical datasets (i.e., ISIC 2020, BRACS, REFUGE) of different modalities and achieves improved results in AUC, BACC, and F1-score evaluation metrics.

II. Related Work

A. Data Imbalance Methods

The current data imbalance solutions predominantly include re-sampling, re-weighting, designing loss functions, and synthesizing data or labels. Re-sampling use over-sampling [4], [23], [24] to increase the number of samples in the minority classes or under-sampling [5], [25], [26] by removing samples for majority classes, which often leads to either over-fitting or under-fitting. Re-weighting adjusts the weighting factor in line with the proportion of samples [7], [27], [28] or adopts weights to the effective number of samples [8]. Some methods introduce new loss functions to change classification boundaries for majority and minority classes [29], [30], and synthesizing data or labels [12]. However, in certain medical contexts, where minority classes are malignant or rare diseases, these methods may have restrictions for medical practices for their diversity and lack of labels [31].

B. Multi-Instance Contrastive Learning

1) Contrastive Learning: Contrastive learning methods, based on self-supervised learning, MoCo [32], SimCLR [33], [34], BYOL [35], and SimSiam [36] transforms unlabeled samples randomly, and result in two augmented views of the same sample as contrastive positive pairs. Among these methods, the core idea of MoCo and SimCLR is to attract the positive pairs and repulse the negative pairs, which benefit from a large number of negative data. BYOL and Simsiam methods use positive sample pairs only as inputs for contrastive learning networks. Furthermore, Li et al. [37] is based on few-shot learning, utilizing a dual-path contrastive learning approach to enhance the acquisition of generalized feature embeddings. It distinguishes classes in natural images using a small number of labeled examples per novel class. Conversely, our anomaly detection based approach utilizes a hierarchical-instance contrastive learning network to facilitate the acquisition of feature embeddings between majority classes, without incorporating malignant samples during training. All of the above methods only use data augmentation to construct positive sample pairs, and their application in medicine is limited.

2) Multi-Instance Contrastive Learning: Different from conventional contrastive learning, Azizi et al. [17] proposed the multi-instance contrastive learning (MICLe) method to deal with unlabeled medical images if multiple images of each patient are available. They treat multiple instance pairs, e.g., different viewpoints or different lighting conditions of the same image, as contrastive medical sample pairs for the pre-training stage. The network learns invariant representation features from not only multiple data transformations of one image but also different viewpoints or conditions of one medical pathology. Vu et al. [38] introduced a MedAug method to use patient metadata as contrastive positive pairs in chest X-rays, so the same patient shares underlying pathologies, hence improving the performance when compared to the ImageNet pre-trained model. These MICLe-based methods are proven to be effective and robust in medical applications. DA-CMIL [39] integrated multi-instances into contrastive learning for chest Computed Tomography (CT). It uses k instances from each patient's CT scan as multi-instance contrastive pairs and validates the method's effectiveness and robustness. However, these methods neglect the hierarchical relationship and only substitute their input in the contrastive learning pre-training model.

C. Anomaly Detection

Anomaly detection is usually used to identify abnormal, novel, or invisible data, and there are GAN-based [19], [21], [22], auto-encoder based [20], [40], [41], knowledge distillation based [42], [43] models, and others [44], [45], [46]. AnoGAN [19] and AnoVAEGAN [20] used the GAN framework, the former was used to detect anomalies of retinal OCT images, and the latter detected MRI brain lesions. In recent anomaly detection methods, the primary focus lies in learning the distribution of anomaly-free data through an auto-encoder network, e.g., RIAD [47], SSM [48]. Most of them learn normal images with good consistency. In the test stage, abnormal images are detected due to their different distributions. However, this series of methods may not be well-suited for handling imbalanced data, as the consistency of samples belonging to the majority class is relatively inadequate when compared to anomaly-free samples. For imbalanced medical data applications, ICOCC [49] method introduces perturbed images generated by given samples, and performs in a better way for learning the inherent features of a given class, since the perturbed imagery can represent complexity and specificity within classes. Nevertheless, they fails to consider the similarity within the class.

III. Methodology

We propose a novel method HCLe to address certain imbalanced medical conditions (e.g., minority classes are malignant and rare diseases). It requires only the majority class to participate in training and learns its distribution during training. When inferring, the minority samples (e.g., melanoma of skin diseases) can be detected for their distribution is different from the majority ones (e.g., benign skin images). With the sub-classes included in the majority, imbalanced cases lead to high variation distributions of intra-class images in the majority class. Therefore, we propose a method with two branches (See Fig. 1) to address these issues: 1) The image reconstruction branch is an auto-encoder framework with multi-constraint functions, which aims to encode good representations to better reconstruct the input image (i.e., images from the majority class), so that the encoded feature embeddings contain satisfactory image-level information for better image reconstruction. 2) The hierarchical-instance contrastive learning branch is a new strategy, which considers intra-class similar representations, i.e., the consistency of intra-class features encoded in the feature embedding. Notably, these two branches share the same encoder and the network is trained end-to-end.

A. Image Reconstruction Branch

Inspired by [47] and [48], we design the image reconstruction branch with multi-constraint functions to extract good representations to reconstruct samples from the majority class. In Fig. 1, first, the input image x is encoded as a feature embedding e by the encoder E. Here it is expected that the encoder E can extract good embeddings from each image, and then a generator G is introduced to reconstruct input x via the encoded embedding e. Finally, we utilize multiple constraint functions to constrain the original image and reconstructed image. The purpose of the encode-decode process is to make the distance between the reconstructed image $\hat{x}$ and the original image x as close as possible.

Instead of traditional $L2$ loss as a constraint function, we not only use a discriminator to improve the feature extraction and reconstruction capabilities of the auto-encoder [31] but also employ structure similarity (SSIM) [50], [51] and gradient magnitude similarity (GMS) [47], [52] losses to extract great image-level representations of the majority class and getting plausible reconstruction results. The SSIM loss pays more attention to global rather than local information [27], which considers luminance, contrast, and anatomical features. GMSD penalizes global variation of local structure between the reconstructed regions and the original image regions. SSIM and GMS are pixel-level similarity metrics focusing on different image properties.

B. Hierarchical-Instance Contrastive Learning Branch

In order to enable the network to consider consistent representations of intra-class samples, we introduce a novel hierarchical-instance contrastive learning branch, which extracts more consistent representations within the majority class and adapts the network to multiple sub-classes per class.

Our hierarchical-instance contrastive learning branch is a novel contrastive learning architecture and has two networks, raw and contrast. The input of raw network is the original input x, and the contrast network input x' is the image randomly sampled from the hierarchical instance of the majority class instead of the image x. Different from data augmentation of the same image (e.g., BYOL [35]) and different viewpoints of the same patient (e.g., MICLe [34]) as contrastive input pairs, this branch's contrastive input pairs are generated from their own sub-classes, which helps adapt contrastive learning to multiple images of the majority class. Such multiple hierarchical instances are available in different sub-classes, patients or images of the same underlying pathology, e.g., nevus and keratosis images of benign skin data. This provides a

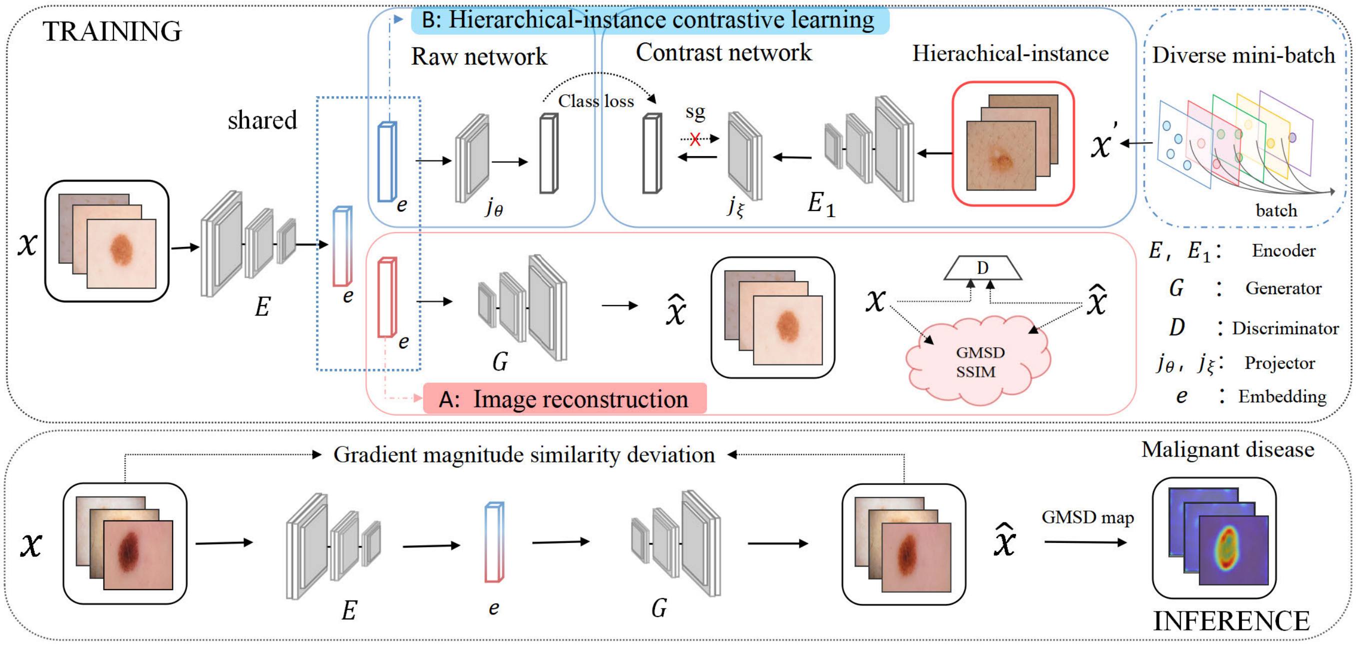

Fig. 1. Our framework consists of two branches, image reconstruction and hierarchical-instance contrastive learning, sharing the same input x and encoder E to learn comprehensive information. The first branch (red box) is to extract good feature representations for image reconstruction using a generator G, and introduces multiple constraints: GMS, SSIM, and adversarial loss (using a discriminator D). The other branch (blue box) takes into account intra-class features, which alleviates the poor consistency of samples in clinical practice. This branch is a new hierarchical-instance contrastive learning strategy containing two networks, raw and contrast. Both the raw network input x and the contrastive input x' are from the majority class but different samples at one time. Branch (B) maintains their instance similarity by minimizing the distance between the outputs of x and x'. These outputs are generated by subjecting x to the encoding process with encoder E followed by projection with projector j_θ, while x' is encoded using encoder E_1 and projected using projector j_ξ, respectively. More importantly, we extend HCLe to a diverse mini-batch version by designing the mini-batch of contrastive inputs. In the inference stage, the feature embedding e of trained encoder E considers both intra-image and intra-class information of the majority class, and then is fed to image reconstruction. A gradient magnitude similarity (GMS) is used to compute the detection map and detection score.

satisfactory contrastive input to learn good intra-class similar features for the contrastive learning network.

Given an image x, it is processed as embedding e via an encoder E, which is shared in the training of two different branches. In Fig. 1 (B), the raw network shows that the embedding e is transformed to z_θ by a projector j_θ. At the same time with the contrast network, the instance image x' sampled from the same training set D, but different from x, e.g., a nevus and a keratosis images of the same benign skin class represent x and x', respectively. Then x' is encoded and projected using encoder E_1 and projector j_ξ. The output vector is denoted as $z_\xi \triangleq j_\xi(E_1(x'))$. The architecture of projector j_θ, j_ξ is a multi-layer perceptron (MLP) [33].

We then use ℓ_2-norm to make z_θ and z_ξ to $\overline{z}_\theta = z_\theta/||z_\theta||_2$ and $\overline{z}_\xi = z_\xi/||z_\xi||_2$. We write the mean squared error between the $\overline{z}_\theta$ and $\overline{z}_\xi$,

$$\mathcal{L}_{\theta,\xi} = ||\overline{z}_\theta - \overline{z}_\xi||_2^2 = ||\overline{I}(x) - \overline{H}(x')||, \tag{1}$$

Here, we denote $I(\cdot) = j_\theta(E(\cdot))$ for the raw network and $H(\cdot) = j_\xi(E_1(\cdot))$ for the contrast network, and exchange the input between the raw and the contrast networks to calculate the symmetrized loss:

$$\mathcal{L}_{class} = ||\overline{I}(x) - \overline{H}(x')|| + ||\overline{I}(x') - \overline{H}(x)|| \tag{2}$$

Moreover, the raw network has weights θ at two stages: an encoder E and projector j_θ. The contrast network has a set of weight ξ, and trains the raw network by offering regression targets [35].

C. Diverse Mini-Batch HCLe Strategy

We further refine the HCLe to diverse mini-batch strategy so that it can perform better in more complex conditions, for example, a more realistic clinical scenario and the identification of the malignant diseases (minority class) from multiple sub-classes of the majority class. For instance, the benign class of the ISIC 2020 skin dataset has most samples that contain multiple sub-classes, e.g., unknown, nevus, solar lentigo, and the model can better identify the minority class (e.g., melanoma) from these benign sub-classes. Moreover, some of these cases are not of clinical significance, e.g., many samples of the benign class can not be identified, which is named unknown class. It does not make sense and is resource-wasting to identify them. Thus they are all treated as the majority class input for model training.

In these cases, for the new diverse mini-batch HCLe strategy, the inputs of the raw network are randomly sampled a mini-batch of N examples and do not need any data augmentation. For the contrast network input x', if multiple sub-classes of the majority class are available (distinguishing these sub-classes are meaningless and resource-wasting), we ensure that each mini-batch N has sampled all sub-classes (see Fig. 2). The proportion of each sub-classes in

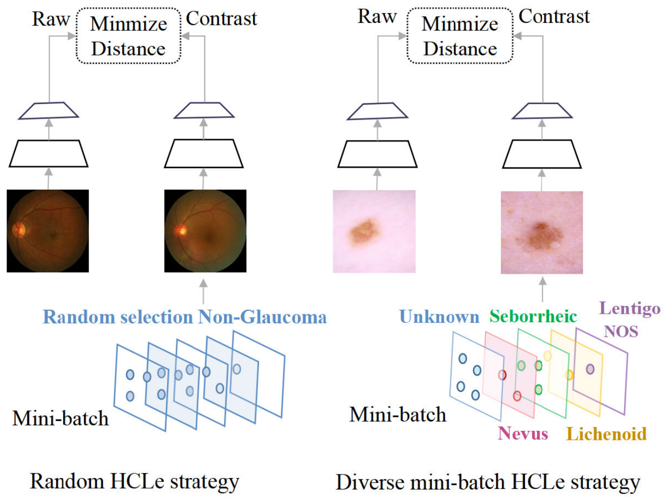

Fig. 2. An illustration of our diverse (right) mini-batch HCLe strategy. If multiple sub-classes of the majority class are available, we design the diverse mini-batch containing every sub-classes as contrast input. Otherwise, we randomly select the mini-batch to the contrast network.

the mini-batch is based on their approximate proportions in the training set. For instance, each mini-batch of ISIC 2020 dataset contains all benign cases, e.g., unknown, nevus, solar lentigo, etc., as contrast inputs. Then the aim is to minimize the distance between representations of the raw network and the contrast network. Each updating step includes all the sub-classes, shortening the distance between the multiple sub-classes within the majority class, thus alleviating the problem caused by poor intra-class consistency. After the training step of each batch size N, the multi-instance self-supervision performs a stochastic optimization step to minimize:

$$\mathcal{L}_{(x_k,x_k')} = \frac{1}{n}\sum_{k=1}^{N}(||\overline{I}(x_k) - \overline{H}(x_k')|| + ||\overline{I}(x_k') - \overline{H}(x_k)||) \quad (3)$$

D. Overall Framework

1) Training Stage: The image reconstruction branch incorporates three constraint functions: gradient magnitude similarity (GMS) [47], [52], structure similarity (SSIM) [50], [51] and adversarial losses. On the other hand, the hierarchical-instance contrastive learning branch penalizes the network using $\mathcal{L}_{class}$.

a) Structure similarity index: Many auto-encoder based methods using L_2 loss (e.g., ℓ^2-Autoencoder) as reconstruction loss cannot reconstruct original images well, nor can they obtain satisfactory anomaly maps since it evaluates each pixel point independently without considering the association of adjacent pixels [51]. Conversely, the structure similarity loss pays more attention to global rather than local information [31], which considers luminance, contrast, and anatomical information [50] to constrain image reconstruction. The structure similarity loss can be defined as:

$$\mathcal{L}_{SSIM} = -SSIM(x, \hat{x}) = -SSIM(x, G(E(x))) \quad (4)$$

where E and G denotes the encoder directly to x and the generator. Moreover, the SSIM index defines a distance measure between image and reconstructed image, considering their similarity in luminance $l(x, \hat{x})$, contrast $c(x, \hat{x})$, and structure $s(x, \hat{x})$:

$$SSIM(x, \hat{x}) = l(x, G(E(x)))^{\alpha} c(x, G(E(x)))^{\beta} s(x, G(E(x)))^{\gamma} \quad (5)$$

where α, β and γ are constants for adjust weights.

b) Gradient magnitude similarity: GMS is unlike L_2 assuming independence between neighboring pixels, it penalizes global variation of local structure between the reconstructed regions and the original image regions. The image gradients are sensitive to image distortions [52], especially the local structures difference between original image and reconstructed image [48], which is computed as follows:

$$g(x) = \sqrt{(x \otimes h_i)^2 + (x \otimes h_j)^2}, \quad (6)$$

where $g(x)$ is the gradient map of original image x. $\otimes$ denotes the convolution operation. h_i and h_j are 3×3 Prewitt filters along the i and j dimensions. The gradient magnitude similarity loss is the mean value of GMS map between image x and reconstructed image $\hat{x}$:

$$\mathcal{L}_{GMS} = -GMS(x, \hat{x}) = -\frac{2g(x)g(\hat{x}) + c}{g(x)^2 + g(\hat{x})^2 + c}, \quad (7)$$

where c is a constant to ensure the stability of the calculation results and to help balance the noise region.

c) Adversarial loss: To further match the distributions of the input, we use PatchGAN [53] to constrain the difference between the original image x and the reconstructed image $\hat{x}$, which can be expressed as:

$$\mathcal{L}_{adv} = \mathbb{E}[log(1 - D_c(\hat{x}))] + \mathbb{E}[log D(x)]. \quad (8)$$

Here, we introduce a discriminator D_c to try to distinguish the fake data (reconstructed image) from a real data (original image).

d) Objective function: Our method consists of two branches. In the hierarchical-instance contrastive learning branch, the input image's embedding is fed into the raw network. Simultaneously, in the image reconstruction branch, the embedding is passed through the generator G of the autoencoder network. Both branches are trained end-to-end. To optimize our method, we employ five loss functions. Therefore, we arrive at the objective function of our model:

$$\mathcal{L} = \lambda_1\mathcal{L}_{SSIM} + \lambda_2\mathcal{L}_{GMS} + \lambda_3\mathcal{L}_{adv} + \lambda_4\mathcal{L}_{class} \quad (9)$$

where λ_1, λ_2, λ_3, λ_4 are the hyper-parameters. After the training step of each batch size N, the network performs a stochastic optimization step to minimize $\mathcal{L}$ of the hierarchical-instance contrastive learning network, and update the weight by a contrast decay rate $\tau \in [0, 1]$. Moreover, a stop-gradient is only applied in contrast network in Fig. 1 (B).

2) Inference Stage: The inference stage is illustrated in Fig. 1. A test image x is fed to the encoder E and generator G, which obtains a reconstructed image $\hat{x}$. Then we calculate the anomaly map by the GMS score map between the original image x and the reconstructed image $\hat{x}$. Since the network never sees the minority class (e.g., malignant diseases) during training, different regions (e.g., malignant regions) may not be reconstructed well. Hence, the distance between the original image and the reconstructed image in the abnormal region may be larger than the normal region. The GMS map function is introduced as follows:

$$\mathcal{M}_(x, \hat{x}) = 1 - GMS(x, \hat{x}) \tag{10}$$

where $\mathcal{M}$ denotes a score function, $GMS(x, \hat{x})$ is GMS maps. To detect minority class, the detection score $\mathcal{A}$ is calculate according to the GMS map of image x and reconstructed image $\hat{x}$. A higher detection score indicates that the image is more likely to be malignant.

IV. Experiments

A. Datasets

Here, three imbalanced underlying diseases with different modalities are compared in this paper.

1) ISIC 2020 [54]: A challenge dataset consisting of 33126 skin appearance images has been compiled for this study, obtained from over 2000 patients. Among these images, only 584 are malignant, whereas the remaining 32542 images have been identified as benign. Nonetheless, it should be noted that a substantial portion of the benign images (over 26000) are classified as “unknown”, indicating that they are indeed benign and yet present difficulties and lack meaningful classification. For the purpose of training our models by employing this dataset, we have utilized various imbalance ratios between malignant and benign samples. Particularly, for the training set, we have utilized ratios of 300:300, 300:1000, 300:2000, and 300:3000 for malignant to benign images, respectively. For the test stage, the benign and malignant images are both 190 from patients who are different from the training set. More importantly, the split of the testing dataset was hierarchical-instance based and designed to simulate complex clinical scenarios in which malignant lesions need to be identified from hierarchical benign samples.

2) BRACS Datasets [55]: The BRACS dataset contains 4539 Regions of Interest (RoIs) from 547 breast carcinoma whole-slide images (WSIs), which H&E stained. These RoIs were classified as normal, benign, Usual ductal hyperplasia (UDH), Atypical Ductal Hyperplasia (ADH), Flat Epithelial Atypia (FEA), Ductal Carcinoma In Situ (DCIS). Distinguishing UDH, FEA, ADH, and DCIS is a big challenge, since UDH, ADH, and DCIS share an intraductal growth pattern, and FEA is easily confused with ADH and DCIS. As a result, non-malignant (UDH, FEA, ADH) and malignant (DCIS) breast images are utilized in this project. Specifically, the aim is to detect malignant (DCIS) breast samples from three non-malignant samples (UDH, FEA, ADH) that share the most similar structural characteristics. Furthermore, the training, validation, and test sets are partitioned using official splits. The dataset exhibits an imbalance ratio of 2.1:1, with the training set comprising a total of 2065 samples and the test set containing 340 samples.

3) REFUGE Datasets [56]: This is a retinal fundus glaucoma challenge dataset with highly imbalanced data for glaucoma detection and optic disc segmentation. In this work, the official splits of this dataset are utilized for fair comparisons. Specifically, this dataset has two distinct types of data imbalances: class imbalance and rare cases. Particularly, the dataset contains a total of 800 samples with clinical ground truth labels, and the training set consists of 400 samples, comprising 360 non-glaucoma samples and 40 glaucoma samples, resulting in an imbalanced ratio of 9:1. Similarly, the test set comprises 400 samples, also demonstrating an imbalanced ratio of 9:1. It should be highlighted that our approach does not require the use of minority samples and relies solely on majority class samples (e.g., 360) for training.

B. Experimental Setup

The model is implemented and developed in Python, PyTorch toolbox. Our models are trained for 800 epochs. The Adam optimizer with an initial learning rate of 0.001 for the first 200 epochs, which is then reduced to 0.0002 until convergence. The input image size is set to 224 × 224. For the ISIC 2020 and BRACS datasets, we set $\lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 = 0.5 : 1 : 1 : 0.5$, and for the REFUGE dataset, we set $\lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 = 1 : 1 : 1 : 0.5$. All networks, including the baseline models, were trained from scratch with a batch-size of 48.

Comparison methods We compare our method with conventional supervised (e.g., re-sampling (RS), focal loss [57], SEDC [13]) and anomaly detection (e.g., VAE-GAN [20], f-AnoGAN [21], RIAD [47]) baselines. For conventional supervised methods, we use ResNet-50 [58], ResNet-101, Vgg16 [59], and DenseNet169 [60] as backbones respectively. In the case of comparison methods, all methods are trained using an initial learning rate of 0.001 for the first 200 epochs, followed by a reduction to 0.0002 for the remaining 600 epochs until convergence. Re-sampling is sampled with a probability that is inversely proportional to the sample size of its class. Similar to our method, VAE-GAN, f-AnoGAN, and RIAD only use the majority class in the training process. RIAD is trained using a varying region size with K = {2, 4, 8, 16.}

C. Evaluation Metrics

In this project, we use three metrics, the area under the receiver operating characteristic curve (AUC), F1-score and balanced accuracy (BACC), to evaluate the classification performance of imbalanced data comprehensively. BACC has been described as the average sum of the sensitivity and specificity, and is more accurate for imbalanced data evaluation than the Accuracy (ACC). The formula is equivalent to the following:

$$BACC = \frac{1}{n} \sum_{k=1}^{n} \frac{TP_k}{TP_k + FN_k} \tag{11}$$

where TP and FN denote true positives and false negatives, and n is the number of classification classes.

V. Results and Discussion

In this section, the performance of our method is compared with the state-of-the-art methods on three different datasets. Furthermore, the effect of different imbalance ratios on these

TABLE I

OUR METHOD IS EVALUATED ON THE THREE DIFFERENT MODALITY DATASETS WITH DIFFERENT IMBALANCE RATIOS

ISIC 2020	Benign					Malignant	Imbalance ratio
	unknown	nevus	Seborrheic	Lentigo NOS	Lichenoid	melanoma	
	2445	445	60	26	24	300	10:1
BRACS	Non-malignant					Malignant	Imbalance ratio
	UDH		FEA	ADH		DCIs	
	389		624	387		665	2.1:1
REFUGE	Non-glaucoma					Glaucoma	Imbalance ratio
	720					80	9:1

TABLE II

QUANTITATIVE COMPARISON OF AUC, BACC, AND F1-SCORE METRICS BETWEEN OUR PROPOSED HCLE AND OTHER METHODS ON ISIC 2020, BRACE AND REFUGE DATASETS. FOR EACH TARGET TASK, WE SHOW THE AVERAGE PERFORMANCE AND STANDARD DEVIATION ACROSS FIVE RUNS. FOR EACH COLUMN, THE TOP VALUE IS HIGHLIGHTED IN BOLD FOR EASY COMPARISON

		Datasets								
		ISIC 2020			BRACS			REFUGE		
Methods	Network	BACC(%)	AUC(%)	F1-score(%)	BACC(%)	AUC(%)	F1-score(%)	BACC(%)	AUC(%)	F1-score(%)
RS	ResNet50	73.34±1.03	75.31±1.29	71.52±0.114	69.21±0.85	71.25±1.11	62.94±0.76	59.31±1.87	59.31±2.59	30.19±2.14
	ResNet101	74.45±1.24	74.64±1.38	72.76±1.18	71.28±0.72	72.10±0.84	67.84±0.82	57.08±3.79	58.06±4.35	26.92±3.62
	Vgg16	70.15±0.95	72.21±1.16	70.24±0.91	67.06±0.98	68.87±1.09	61.64±0.64	74.58±1.56	69.17±1.84	53.47±1.24
	DenseNet169	71.89±1.27	71.42±1.46	72.26±1.03	67.64±1.05	71.09±1.07	58.20±0.86	61.96±2.18	61.39±2.87	26.56±2.04
Focal loss [57]	ResNet50	62.56±0.84	63.25±0.90	53.34±0.62	72.16±0.95	79.64±1.16	68.21±0.79	64.92±2.54	66.87±2.85	40.87±3.96
	ResNet101	62.58±0.96	64.47±1.39	43.36±0.79	69.49±1.32	80.87±1.37	64.75±0.96	50.03±1.22	58.82±1.40	9.62±4.74
	Vgg16	65.81±1.51	67.02±1.86	54.97±1.06	66.56±0.85	58.78±0.79	59.69±0.97	53.07±2.82	54.15±3.26	11.78±2.26
	DenseNet169	62.36±1.97	64.42±1.39	40.49±1.14	61.20±0.96	71.87±0.97	51.71±0.76	63.31±3.32	68.55±3.41	26.27±2.74
SEDC [13]		-	-	-	-	-	-	**88.47**	93.32	-
VAE-GAN [20]		68.32±1.69	79.62±2.13	68.26±1.04	66.76±0.96	70.21±0.76	70.19±0.67	71.21±3.16	77.94±3.59	60.83±3.92
f-AnoGAN [21]		73.81±1.26	80.67±1.37	73.24±0.86	71.36±1.03	74.61±1.28	72.41±0.82	65.97±2.69	78.68±2.42	36.67±2.27
RIAD [47]		76.12±0.78	81.17±0.86	76.02±0.62	71.32±0.95	7741±0.86	71.24±0.63	67.86±1.27	82.21±1.49	57.02±0.97
HCLe		**82.76±0.94**	**86.83±1.16**	**82.97±0.76**	**76.57±0.92**	**82.63±1.14**	**77.16±0.81**	87.36±2.72	**96.26±1.61**	**74.76±2.53**

methods is evaluated. Finally, comprehensive ablation studies are conducted to justify the rationality of the design of the proposed HCLe.

A. Comparison Performance on Different Datasets

1) Results on ISIC 2020 Dataset: The performance of our method, traditional imbalanced methods, and anomaly detection methods on the ISIC 2020 dataset is reported in TABLE II, and the imbalance ratio is extremely high at 10:1 (in TABLE I). It can be seen from TABLE II that our method achieves the best performances on BACC, AUC, and F1-score compared to conventional supervised or anomaly detection methods. Moreover, it increases the BACC, AUC, and F1-score by 6.64%, 5.66%, and 6.95% respectively, compared to the second-best baseline. This indicates that our method is effective for the skin ISIC 2020 dataset with interwoven imbalances, more serious imbalance ratio (10:1) and high intra-class variation.

2) Results on BRACS Dataset: To validate the model's performance on different medical diseases, we also conduct in-depth experiments on the breast pathology slides, which have an inhomogeneous appearance and are coupled with data imbalance. Moreover, the TABLE II indicates that our method achieves the best performance on three evaluation metrics, 76.57% of BACC, 82.63% of AUC, and 77.16% of F1-score. Particularly in BACC and F1-score, they are 4.41% and 4.75% higher than the second-highest score, respectively, which demonstrates that our method HCLe can cope with imbalanced breast pathological data well.

3) Results on REFUGE Dataset: The performance of our method and state-of-the-art approaches on glaucoma diseases is finally evaluated, and glaucoma is also a disease with a highly imbalanced number of cases. Moreover, REFUGE is a glaucoma fundus dataset with an imbalance rate of 9 and a small number of samples, including training and testing sets. Our model demonstrates great performance, with scores of 87.36%, 96.26%, and 74.76% on BACC, AUC, and F1 scores, respectively. Furthermore, it outperforms all other methods, securing the top rank in both AUC and F1-score metrics.

4) Results of Different Imbalance Ratios on the ISIC 2020 Dataset: To further validate the effect of our method with different imbalance ratios, we conduct more experiments on the ISIS2020 dataset. The performance of our method is compared with that of conventional and anomaly detection methods, and experiments are designed, starting with a 1:1 balanced ratio. The TABLE III (column 300:300 (1:1)) and Fig 3 illustrate that the conventional method outperforms the anomaly detection methods in all three evaluation metrics with a balanced ratio (1 : 1). Nonetheless, by increasing the imbalance ratio to 3.3 : 1, the performance of the conventional methods rises slowly. Moreover, on the condition that the imbalance ratio becomes 6.7 : 1 or higher, the performance of the conventional method starts to drop significantly. On the contrary, the anomaly detection method indicates an upward trend from increasing the imbalance ratio from 1 : 1 to 3.3 : 1 and 6.7 : 1. Notably, our method outperforms the conventional method with an imbalance ratio of 3.3 : 1 or higher.

B. Ablation Study

1) Effectiveness of Sharing Strategy: For the purpose of evaluating the contribution of each component in our method, we conduct experiments revealed in TABLE IV. 'Image-level' and 'Intra-class' denote image reconstruction branch and hierarchical-instance contrastive learning branch. For the 'Intra-class' alone, the detection score to detect the minority

TABLE III

PERFORMANCE OF OUR HCLE METHOD IN COMPARISON TO OTHER EXISTING METHODS, USING THREE DIFFERENT IMBALANCE RATIOS (300:300, 1000:300, 2000:300) ON THE ISIC 2020 DATASET. FOR EACH TARGET TASK, WE SHOW THE AVERAGE PERFORMANCE AND STANDARD DEVIATION ACROSS FIVE RUNS. FOR EACH COLUMN, THE TOP VALUE IS HIGHLIGHTED IN BOLD FOR EASY COMPARISON

		Imbalance ratio								
		300 : 300 (1 : 1)			1000 : 300 (3.3 : 1)			2000 : 300 (6.7 : 1)		
Methods	Network	BACC(%)	AUC(%)	F1-score(%)	BACC(%)	AUC(%)	F1-score(%)	BACC(%)	AUC(%)	F1-score(%)
RS	ResNet50	74.72±0.68	75.26±0.71	72.74±0.63	77.68±0.42	78.71±0.79	75.43±0.58	74.73±1.64	75.67±1.19	73.03±1.35
	ResNet101	**77.29±0.91**	**77.22±0.96**	**76.16±0.96**	77.14±0.57	78.73±0.79	73.81±0.51	71.24±1.20	72.91±1.36	65.94±1.08
	Vgg16	72.71±1.16	74.81±1.43	67.81±1.47	76.78±0.71	77.52±0.52	74.59±0.61	70.67±1.56	72.82±1.49	70.39±1.41
	DenseNet169	75.91±0.90	76.12±1.12	75.51±0.93	74.55±1.53	74.86±1.05	75.26±0.82	75.32±0.84	76.04±0.71	78.81±0.96
Focal loss [57]	ResNet50	74.86±0.82	76.89±0.61	74.53±0.67	71.12±0.63	74.86±0.86	61.51±0.89	63.42±1.62	66.97±1.71	61.37±1.24
	ResNet101	72.18±1.26	73.56±1.17	69.32±1.06	70.92±0.96	71.26±0.74	62.72±0.93	63.86±2.68	64.11±2.34	45.21±2.64
	Vgg16	74.16±1.11	75.23±0.74	74.03±1.04	70.39±1.28	72.84±1.16	69.76±1.31	68.04±1.92	68.96±1.36	58.86±1.98
	DenseNet169	73.56±0.96	73.13±0.86	74.68±0.92	63.06±2.96	63.81±2.36	46.97±2.97	55.71±1.85	56.58±1.74	21.06±1.92
VAE-GAN [20]		68.11±1.63	76.06±1.24	65.31±1.36	69.64±1.12	77.43±1.28	68.26±0.96	67.78±1.16	78.41±0.96	66.97±1.03
f-AnoGAN [21]		63.76±1.71	71.86±1.51	63.57±1.46	67.89±1.26	75.16±0.93	66.58±1.15	72.26±0.72	79.62±0.82	72.49±0.67
RIAD [47]		70.59±1.74	75.36±1.61	71.21±1.27	71.39±1.33	76.47±1.39	71.42±1.21	75.26±0.86	78.73±0.92	73.22±0.74
HCLe		72.27±1.48	76.74±1.32	72.83±1.26	**78.54±1.16**	**81.93±1.09**	**77.92±1.04**	**81.31±1.06**	**85.39±0.98**	**80.76±0.93**

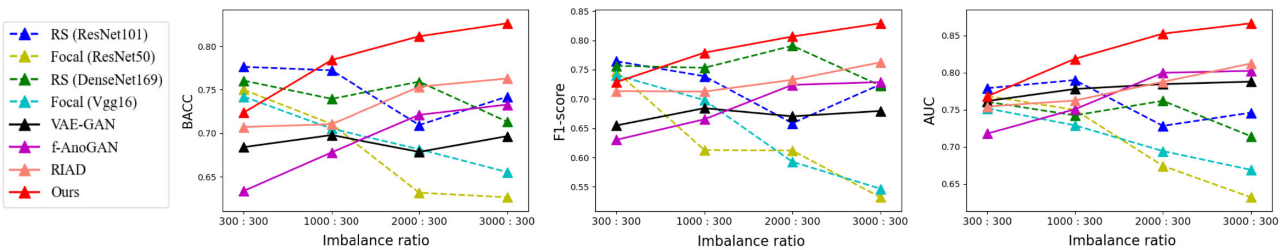

Fig. 3. The trend of different balance ratios in the three evaluation indicators with conventional and anomaly detection methods.

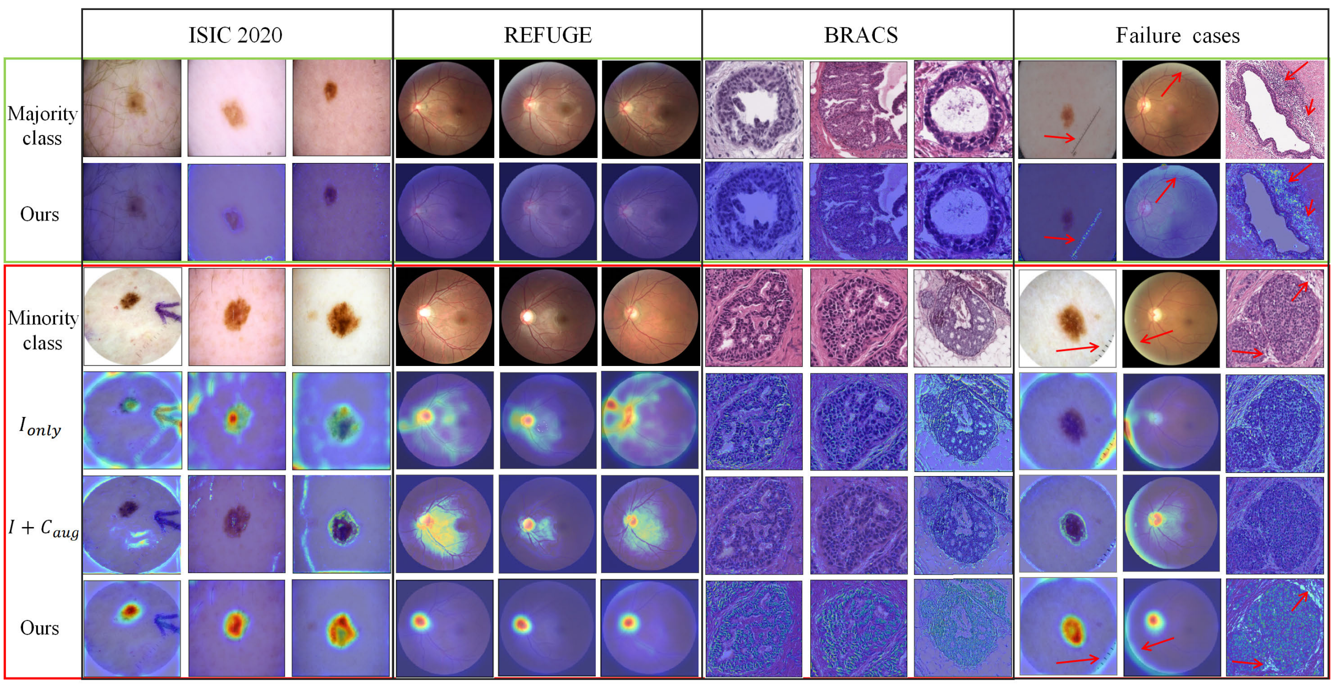

Fig. 4. Unsupervised pixel-level detection results of our proposed method. The green box indicates the majority class's test images and anomaly maps obtained by our method. The red box (the top raw) represents test images from the minority class, and 'I_{only}' is anomaly maps obtained only by the image reconstruction branch. '$I + C_{aug}$' presents two branches that share the same encoder E and use the data augmentation strategy. The last row of the red box is anomaly maps from our method. The results of the three datasets (i.e., ISIC 2020, REFUGE, BRACS) demonstrate the effectiveness of our approach in accurately detecting malignant regions. However, from the first column of failure cases, it indicates occasional misclassification of clinical markings regions (highlighted by the red arrows) as malignant regions by the proposed framework. Similarly, the last two columns of the failure cases show that our method occasionally misclassifies regions with altered edge brightness (highlighted by the red arrows) as anomalies.

class was obtained by adopting the output from the E_x in Fig. 1. In the case of 'share E', it refers to 'image-level' and 'intra-class' branches sharing the same encoder E. When these two branches do not share the encoder E, the detection score is calculated by summing score maps from the two branches.

TABLE IV
ABLATION STUDIES FOR FRAMEWORK DESIGN ON THREE DATASETS. 'IMAGE-LEVEL' AND 'INTRA-CLASS' DENOTE IMAGE RECONSTRUCTION AND HIERARCHICAL-INSTANCE CONTRASTIVE LEARNING BRANCHES. 'SHARE E' MEANS 'IMAGE-LEVEL' AND 'INTRA-CLASS' BRANCHES SHARING THE SAME ENCODER E; OTHERWISE, THESE TWO BRANCHES ARE NOT SHARED. FOR EACH TASK, THE TOP VALUES ARE HIGHLIGHTED IN BOLD FOR EASY COMPARISON

Datasets	Strategies			BACC	AUC	F1-score
	Image-level	Intra-class	Share E			
ISIC 2020	✓			0.6842	0.6946	0.6429
		✓		0.6911	0.6808	0.6651
	✓	✓		0.7474	0.7692	0.7397
	✓	✓	✓	**0.8276**	**0.8683**	**0.8297**
BRACS	✓			0.6623	0.6949	0.6463
		✓		0.6112	0.6438	0.6298
	✓	✓		0.7090	0.7435	0.7023
	✓	✓	✓	**0.7657**	**0.8263**	**0.7716**
REFUGE	✓			0.5819	0.6629	0.3218
		✓		0.6291	0.7420	0.4889
	✓	✓		0.7624	0.8115	0.5439
	✓	✓	✓	**0.8736**	**0.9626**	**0.7476**

The results are indicated in TABLE IV, where both the 'Image-level' alone and the 'Intra-class' alone achieve extremely limited performance. However, training with the concatenation of the two branches results in higher performance. Furthermore, the shared encoder E substantially strengthened the performance, as evidenced by the improvements in BACC, AUC, and F1-score for the ISIC 2020 dataset, which are 0.0789, 0.0973, and 0.0893, respectively. This conclusion is also validated in the other two datasets, indicating the effectiveness of the sharing strategy.

2) Effectiveness of Diverse Mini-Batch Strategy in HCLe: Here, we compare three strategies i.e., data augmentation, Random HCLe, and Diverse HCLe. For the data augmentation strategy, widely adopted techniques such as Flip, Gaussian blur, and Color jitter are applied to the inputs x and x' of the raw network and contrast network. Besides, the remaining experimental settings are kept the same for a fair comparison. Furthermore, two additional HCLe strategies are compared: 1) Random HCLe: randomly select contrastive pairs from hierarchical instances of the majority class, i.e., the contrastive network input x' random selected from the training set. 2) Diverse HCLe: For the contrastive network input x', we sample each mini-batch of N samples containing all sub-classes of a majority class according to their approximate proportions in the training set.

As can be seen from TABLE V, compared with the data augmentation strategy, HCLe strategies achieve superior performance, which indicates HCLe strategies can leverage unlabeled data of the training set to improve the performance significantly in three datasets. Furthermore, if multiple sub-classes are available, the diverse HCLe strategy can steadily improve the minority detection performance (e.g., ISIC 2020 dataset from 0.8024, 0.8346 and 0.7991 to 0.8263, 0.8665 and 0.8290 in BACC, AUC and F1-score). In the REFUGE dataset, where a sub-class is lacking in the majority class, both the random and diverse HCLe strategies yield the same results. These quantitative results demonstrate the effectiveness of the diverse HCLe strategy in the prediction task, as it enables more accurate detection of malignant diseases (the minority class), and cannot be substituted by data augmentation or random strategies.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT STRATEGIES OF THE HCLE BRANCH ON THREE DATASETS. 'AUGMENTATION' INDICATES THE USE OF TRADITIONAL DATA AUGMENTATION TECHNIQUES IN THE HCLE NETWORK. 'RANDOM HCLE' REPRESENTS CONTRASTIVE SAMPLES SELECTED ENTIRELY AT RANDOM. 'DIVERSE HCLE' MEANS DIVERSE MINI-BATCH HCLE STRATEGY, WHICH ENSURES THAT EACH MINI-BATCH HAS SAMPLED ALL SUB-CLASSES. FOR EACH TASK, THE TOP VALUES ARE HIGHLIGHTED IN BOLD FOR EASY COMPARISON

Dataset	Strategies	BACC	AUC	F1-score
ISIC 2020	Augmentation	0.7662	0.7846	0.7696
	Random HCLe	0.8024	0.8346	0.7991
	Diverse HCLe	**0.8276**	**0.8683**	**0.8297**
BRACS	Augmentation	0.7176	0.7616	0.7285
	Random HCLe	0.7487	0.8016	0.7684
	Diverse HCLe	**0.7657**	**0.8263**	**0.7716**
REFUGE	Augmentation	0.8278	0.9099	0.5690
	Random HCLe	**0.8736**	**0.9626**	**0.7476**
	Diverse HCLe	**0.8736**	**0.9626**	**0.7476**

TABLE VI
PERFORMANCE COMPARISON OF THREE IMAGE-RECONSTRUCTION CONSTRAINT FUNCTIONS IN THE HCLE METHOD ON THREE DATASETS. FOR EACH TASK, THE TOP VALUES ARE HIGHLIGHTED IN BOLD FOR EASY COMPARISON

Dataset	Functions	BACC	AUC	F1-score
ISIC 2020	$\mathcal{L}_{adv}$	0.7564	0.7621	0.7403
	$\mathcal{L}_{adv}+\mathcal{L}_{GMS}$	0.8026	0.8349	0.8158
	$\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	0.7763	0.7832	0.7694
	$\mathcal{L}_{adv}+\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	**0.8276**	**0.8683**	**0.8297**
BRACS	$\mathcal{L}_{adv}$	0.7013	0.7134	0.7023
	$\mathcal{L}_{adv}+\mathcal{L}_{GMS}$	**0.7667**	0.8046	0.7553
	$\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	0.7183	0.7426	0.7229
	$\mathcal{L}_{adv}+\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	0.7657	**0.8263**	**0.7716**
REFUGE	$\mathcal{L}_{adv}$	0.7042	0.8593	0.4578
	$\mathcal{L}_{adv}+\mathcal{L}_{GMS}$	0.8320	**0.9771**	0.7406
	$\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	0.7764	0.9024	0.5495
	$\mathcal{L}_{adv}+\mathcal{L}_{SSIM}+\mathcal{L}_{GMS}$	**0.8736**	0.9626	**0.7476**

3) Effectiveness of Image Reconstruction Constraint Functions: TABLE VI compares the effectiveness of these constrain functions. When using multiple constraint functions ($\mathcal{L}_{adv}$, $\mathcal{L}_{SSIM}$, and $\mathcal{L}_{GMS}$ loss), the model performs best on all metrics in ISIC 2020 datasets, and it performs best on two metrics in BRACS and REFUGE datasets. Specifically, our multiple constraints increase the BACC, AUC, and F1-score in ISIC 2020 dataset by 0.0237, 0.0316, and 0.0132 respectively, compared to the second-best baseline. TABLE VI confirms that multiple constraint functions are effective for medical images of different diseases and modalities.

C. Significant Difference Analysis

We conducted a comprehensive assessment to analyze the statistical significance of the achieved performance improvement. Specifically, the statistical significance of the proposed HCLe versus other methods including baseline approaches and alternative strategies of our method, is examined by the paired t-test. A p-value less than 0.05 indicates that the method performs significantly better than the baseline method. Based on the results presented in the TABLE VII, it demonstrates that all improvements are statistically significant. These results further demonstrate the effectiveness of the proposed HCLe.

TABLE VII

STATISTICAL SIGNIFICANCE (p-VALUE) OF THE PROPOSED HCLE COMPARED WITH OTHER METHODS (I.E., BASELINE METHODS AND OTHER STRATEGIES OF OUR APPROACH). THE 'I_{only}' IS ANOMALY MAPS OBTAINED ONLY BY THE IMAGE RECONSTRUCTION BRANCH. '$I + C_{aug}$' PRESENTS TWO BRANCHES THAT SHARE THE SAME ENCODER E AND USE THE DATA AUGMENTATION STRATEGY

Network	ISIC2020	BRACS	REFUGE
HCLe / RS_{Res101}	0.015	0.011	<5e-4
HCLe / RS_{vgg16}	0.008	0.006	0.001
HCLe / $Focal_{Res101}$	0.003	0.010	<5e-4
HCLe / $Focal_{vgg16}$	0.004	0.007	<5e-4
HCLe / VAE-GAN	0.016	0.008	0.006
HCLe / f-AnoGAN	0.027	0.031	0.002
HCLe / RIAD	0.044	0.037	0.003
HCLe / I_{only}	0.001	<5e-4	<5e-4
HCLe / $(I + C_{aug})$	0.042	0.035	0.028

VI. CONCLUSION

In this work, a novel approach was proposed to address imbalanced conditions in certain medical diseases. The training stage only requires the majority class (e.g., normal or benign data) to participate in the training stage, and the minority class is detected because it is out of distribution in the inference stage. To enhance the learning of the majority class distribution, improving the feature extraction ability, and addressing the inconsistency issue were deemed necessary, our method employed two branches, namely image reconstruction and hierarchical-instance contrastive learning. The former incorporated multi-constraint functions into the auto-encoder network to facilitate better feature extraction. Meanwhile, the latter introduced a new hierarchical-instance contrastive learning strategy, which aimed to enhance the consistency of feature representation among majority samples through multi-instance contrastive learning.

Subsequently, we evaluate our method on three medical datasets (i.e., ISIC 2020, BRACS, REFUGE) with different modalities. The experimental results demonstrate that the proposed method achieves improved performance in AUC, BACC, and F1-score evaluation metrics compared with conventional and anomaly detection methods. On three datasets, the HCLe strategy improves classification performance relative to the data augmentation strategy. Moreover, according to the HCLe, the diverse mini-batch strategy performs better than the random mini-batch strategy.

Moreover, due to the generality of the proposed HCLe method, the HCLe can be further extended to other imbalanced medical image tasks and even try anomaly segmentation of these imbalanced images. Although our method achieves great performance, it still comes with some unresolved limitations. For instance, it may not effectively detect certain malignant or rare diseases in cases where the lesion area closely resembles the benign tissue area, as their feature distributions exhibit no obvious differences. In light of this, future research will prioritize the segmentation and detection of these challenging diseases.

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